

Introduction to Quantum Computing

Key Images: Week 3

Quantum Algorithm Speed-Ups > Quantum Advantage

QUANTUM ALGORITHM SPEEDUPS

ALGORITHM	CLASSICAL RESOURCES	QUANTUM RESOURCES	QUANTUM ADVANTAGE	LIMITATION
Simulation (quantum chemistry)	2^N (for N atoms)	N^C	Exponential*	Mapping problem to qubits
Factoring (+ related number theoretic)	2^N (for N digits)	N^3	Exponential	Classical runtime limit unproven
Linear systems ($Ax=b$)	2^N (for N digits)	$\sim N$	Exponential	Strict conditions, e.g. sparse matrix
Optimization	2^N (for N items)	?	?	Empirical
Search (unsorted/unstructured data)	N (for N entries)	$\sqrt{N}$	Polynomial ($\sqrt{N}$)	Data loading